

■ ANCHOR: DISTRACTION-FREE INSTAGRAM

Product Design Report & Engineering Documentation (PDR)

Package: com.anchor.app | **Platforms:** Android (React Native 0.86 / Expo SDK 57) & macOS AppKit Simulator | **Updated:** September 2026

1. Executive Summary & Mission

Modern social media applications are engineered with algorithmic recommendation grids, infinite video feeds (Reels/Explore), and engagement-maximizing feedback loops that foster compulsive consumption. **Anchor** is a minimalist, distraction-free Instagram client engineered to eliminate passive content consumption while strictly preserving active interpersonal communication.

- **Eradicated:** Reels feeds (/reels/), Explore recommendation grids (/explore/), Home feed posts, 'Suggested for you' carousels, and persistent 'Get the app' banners.
- **Preserved:** 1-on-1 Direct Messages, group chats, Notes, media attachments, Activity notifications (likes, comments, follow requests), Personal Profile management, and Instagram Settings.

2. System Architecture & Tech Matrix

Anchor features a hybrid native-web architecture. The outer shell provides native status bars, safe-area padding, and touch-optimized navigation controls. The inner viewport utilizes a hardware-accelerated WebView with custom script/style injection engines that surgically shape the DOM before rendering.

Layer	Android Mobile App	macOS Desktop Simulator
Runtime	React Native 0.86.3 / Expo SDK 57	macOS AppKit (Swift / NSApplication)
Rendering Engine	Android System WebView (Chromium)	Apple WebKit (WKWebView)
Acceleration	androidLayerType='hardware', GPU texture	Metal Hardware Accelerated Compositor
Navigation Shell	React Native Inset-Aware Stack View	Native NSScrollView / Auto Layout
Build Pipeline	EAS Build (APK Preview / Production)	swiftc desktop_runner.swift -o AnchorDesktop
Color System	OLED Black #000000, Emerald #0fb981	OLED Black #000000, Emerald #0fb981

3. Core UI Components & Navigation Shell

3.1. Anchor Top Navigation Bar

- **Brand Title & Status:** Renders ■ Anchor alongside an emerald ● Focused pill indicator.
- **Smart Back Navigation (■):** Detects in-page Instagram back buttons (e.g. ■ or ←), falls back to SPA history pop (window.history.back()), and ultimately calls native webView.goBack().
- **Hard Restart (■):** Resets active trigger states and forces an un-cached reload of the active page.
- **Ergonomics:** Expanded 36x36dp touch targets with 12dp hitSlop and tactile activeOpacity feedback.

3.2. Anchor 4-Tab Bottom Navigation Bar

- **Activity Tab:** Opens notification drawer without triggering feed redirects or collapsing items.
- **Messages Tab:** Default landing view loading Direct Inbox (/direct/inbox/).
- **Profile Tab:** Dynamically resolves active session to open the logged-in profile (/username/).
- **Settings Tab:** Routes directly to Instagram settings (/accounts/settings/) without modal popups.
- **Gesture Inset Protection:** Integrated useSafeAreaInsets() so labels are never covered by Android's navigation bar.

4. Key Engineering Inventions & DOM Engines

4.1. Non-Destructive Geometric Bottom Bar Filter

Instagram's mobile web framework renders a redundant bottom bar (Direct plane + avatar). Earlier attempts using CSS selectors (nav, [role='navigation']) hid the direct chat list, producing completely blank screens. Anchor solved this with a **geometric filter** targeting elements based strictly on viewport metrics:

Filter Condition: *position: fixed/sticky AND height $\in [35, 85]px$ AND top $\geq vh - 95px$ AND width $\geq 70\% vw$ AND hasNoInput(el).*

This completely isolates and hides Instagram's floating web bar while leaving 700px+ chat containers and composer inputs 100% intact.

4.2. Dynamic Multi-Tier User Detection (Zero Hardcoding)

Previous prototypes defaulted to hardcoded usernames (e.g. jack_98472). Anchor now uses an autonomous 5-tier session resolver (detectLoggedInUser):

1. **Top Bar Account Switcher:** Inspects heading elements in the top 90px containing downward chevrons (nagumoo_001 ▼).
2. **Avatar DOM Links:** Extracts profile link paths matching `a[href^='/'] img[alt*='profile picture']`.
3. **Image Alt Metadata:** Evaluates regex `/^("["]+)"s profile picture/i`.
4. **JavaScript Viewer Globals:** Inspects `window._sharedData.config.viewer.username` and `window.__initialData`.
5. **Polaris LocalStorage:** Scans session keys for validated active username strings.

4.3. Anti-Upsell & App Banner Purger

Permanent suppression of 'Get the app' and 'Use the app' prompts via dual-layer enforcement: CSS preloading blocks download anchors, while a leaf-node text scanner identifies and disables prompt banners without affecting main column containers.

5. Problem-Solving & Verification History

Issue Observed	Underlying Root Cause	Permanent Engineering Solution
Profile opened wrong user (jack_98472)	Hardcoded handle cached in Hermes bytecode dist bundle.	Purged all hardcoded handles; implemented dynamic 5-tier detectLoggedInUser().
Blank Screen on Android	CSS nav, [role='navigation'] and JS ancestor crawl hid chat container.	Removed destructive rules; deployed non-destructive geometric filtering.
Empty Notification Center	Rule header ~ div:has(...) hid page body; article rule hid notifications; heart clicked in loop.	Removed header ~ div and article rules; added single-fire guard __ACTIVITY_TRIGGERED__.
Gesture Bar Overlap	Bottom navigation bar had static 56dp height with zero bottom padding.	Integrated useSafeAreaInsets() to add dynamic paddingBottom matching device gesture bar.
Top Navbar Missing	Top header was present only in desktop runner, not ported to mobile App.js.	Built complete topHeader in React Native with brand title, status dot, and controls.
Back Button Inoperative	React Native canGoBack returns false for HTML5 pushState navigations.	Multi-level back handler: triggers Instagram in-page chevron, SPA history, then native WebView.

6. Deployment & Build Instructions

Updating Installed Android APK (In-Place Update):

1. Commit code: `git add . && git commit -m 'Release notes'`
2. Build via EAS: `npx eas build -p android --profile preview`
3. Download APK and tap **Update** on Android (keeps login cookies and data intact).

Live Testing via Expo Go:

Run `npx expo start` and scan the QR code using Expo Go on Android for instant live reloading.

Compiling macOS Desktop Simulator:

Run `swiftc desktop_runner.swift -o AnchorDesktop && ./run_desktop.sh`

Project Status: Production Ready & Verified | **Author:** Ishant Verma | **Anchor Engineering Team**